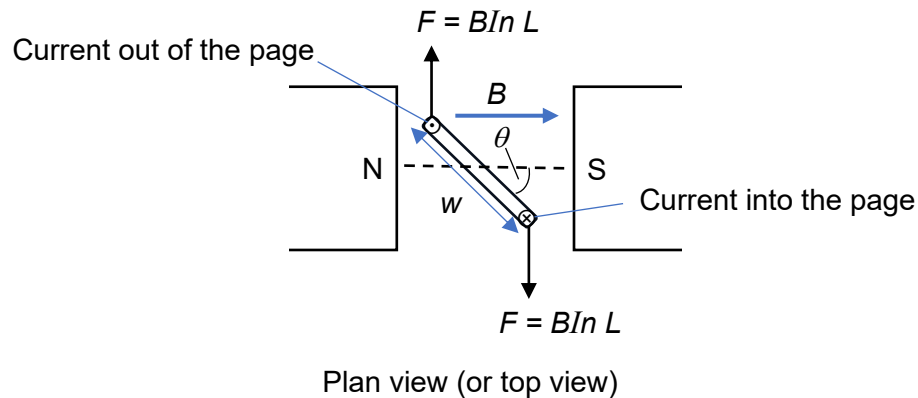
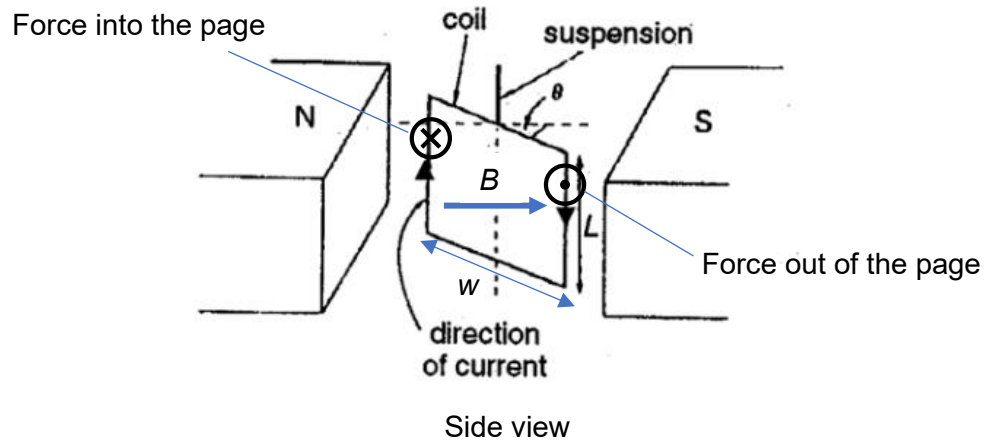


24

D

The forces acting on the vertical sides of the coil = $nBIL$, since the current in the vertical wires are always perpendicular to the horizontal magnetic field.



The angle is required only for the perpendicular distance between the lines of action of the forces $w \cos \theta$.

The moment of the couple is $F \times d_{\perp} = nBIL w \cos \theta$.

25

D

Take the upper side of the coil as our reference area